

Forecasting Tourist Arrivals to Sri Lanka

Random Forest Regression Using Historical Tourist Arrival Data

Machine Learning | Regression

PROBLEM

Tourist arrivals vary across countries and months, making future tourism demand difficult to estimate.

OBJECTIVE

To predict monthly tourist arrivals to Sri Lanka using historical country-level arrival patterns.

DATASET

SLTDA -Tourist Arrivals from All Countries

Sri Lanka Tourism Development Authority (SLTDA) (2018 – Sep. 2026)

20,072 country-month observations

Target Variable : Tourist Arrivals

SOURCE

MACHINE LEARNING APPROACH

Task:

Regression ————— Random Forest Regressor

Model:



Captures non-linear relationships between country, time and historical arrival patterns.

TRAIN / VALIDATION / TEST

TRAIN
2019–2024

VALIDATION
2025

TEST
Jan–Sep 2026

Chronological splitting prevents future observations from being used to train the model.

MODEL RESULTS

2026 Test Set

R²
0.920
Variation explained higher is better

MAE
288.36
Average absolute error 288 tourists per country

RMSE
1,216.68
Penalized larger errors more sensitive to large misses.

PREPROCESSING

CLEAN

Standardized country names

1

2

ORGANIZE

Constructed chronological dates

CHECK

Checked missing values and duplicates

3

4

ENGINEER FEATURES

Created historical arrival features

Features Used

- Previous-month arrivals
- Same-month previous-year arrivals
- 3-month rolling average

These features capture recent and seasonal arrival patterns.

FUTURE FORECAST

Predicted Annual Tourist Arrivals to Sri Lanka



* 2026 forecast covers October – December only, so it is not shown against full years.

The Random Forest model achieved an R² of 0.920 on unseen 2026 test data, showing that it captured much of the variation in monthly country-level tourist arrivals.

LIMITATIONS

- Data is monthly, so daily arrivals are not predicted.
- Long-range forecasts become increasingly uncertain.
- External factors such as economic conditions, policy changes and unexpected disruptions are not explicitly modeled.